The propositions package*

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Abstract

The propositions package provides a key-value driven system for labelling propositions, theses, and premises in academic papers. Items may be given names like '(P)' or 'Physicalism', or auto-numbered using different counters; all carry robust cross-references with configurable formatting. The package integrates with amsmath, hyperref, and cleveref.

Contents

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1 Introduction

In some academic disciplines (such as philosophy), it is common to have displayed propositions (examples, theses, premises,. . .) with various kinds of labels. A thesis might be referred to as '(P)' or 'Physicalism'; the premises of an argument might be numbered as 'P1', 'P2', 'P3', . . . ; or examples might be numbered consecutively over the course of a whole article. Standard LaTeX environments like `enumerate` can handle some of these cases, but cross-referencing is awkward: `\ref` produces a bare number or letter, and the author must manually add parentheses or other formatting at every point of reference. The standard `description` environment does not allow cross-referencing at all.

The propositions package solves this by attaching formatting information to each label. A short item like `\pitem[P]` is displayed as "(P)" and `\ref` automatically produces "(P)" as well—complete with parentheses, hyperlinks, and cleveref support. The full key-value interface supports named items, numbered items, custom counters, glosses, shorthands, and per-item format overrides.

The `\ptag` command (which requires amsmath) extends this to displayed math environments: an equation can be tagged with a proposition label instead of (or using) its equation number.

2 History

I wrote the ancestor to this package in the 90s while finishing my Ph.D. thesis, but never documented it or shared it with the world. This new version is a thorough re-implementation in LaTeX3, written in 2026 with extensive help from Claude Code. I hope others will find it as useful as I have.

3 Basic usage

Load the package with `\usepackage{propositions}` or `\usepackage[⟨options⟩]{propositions}` (see section 8 below for valid package options).

The `prop` environment generates a list of propositions, each introduced by a `\pitem`. `\pitem` with an optional argument gives a `description`-like label:

```
\begin{prop}
  \pitem[Physicalism] Everything is
  physical. \label{phys}
  \pitem[Idealism] Everything is
  mental. \label{ideal}
\end{prop}
```

Physicalism Everything is physical.

Idealism Everything is mental.

Unlike the standard `description` environment, one can refer back to these propositions using the standard `\ref` command:

```
\ref{phys} is more plausible
than \ref{ideal}.
```

Physicalism is more plausible than **Idealism**.

With no optional argument, `\pitem` will by default generate numbered items similar to `enumerate`, but with numbering that persists across the document:

```
\begin{prop}
  \pitem Every atom is physical.
  \label{atoms}
\end{prop}
\ref{phys} follows from the
conjunction of \ref{atoms} and
\begin{prop}
  \pitem Everything is an atom.
  \label{atomism}
\end{prop}
```

(1) Every atom is physical.

Physicalism follows from the conjunction of (1) and

(2) Everything is an atom.

As with `enumerate`, the counter and formatting depend on the nesting level:

```
\begin{prop}
  \pitem \label{dual}
  \begin{prop}
    \pitem Some things are physical.
    \label{some}
    \pitem Some things are not
    physical. \label{notall}
  \end{prop}
\end{prop}
Without \ref{some}, \ref{dual}
would be consistent with \ref{ideal}.
```

(3) a. Some things are physical.

b. Some things are not physical.

Without (3a), (3) would be consistent with **Idealism**.

4 Advanced usage

The format of the proposition labels, and of subsequent references, are both configurable using a key=value syntax (see section 5 for the possible keys):

```
\begin{prop}
  \pitem[No Overlap,
    align=flush,
    display format=\textbf{#1},
    ref format=\textit{#1}]
    Nothing mental is
    physical. \label{incomp}
\end{prop}
Is \ref{dual} consistent
with \ref{phys}?
```

No Overlap Nothing mental is physical.

Is (3) consistent with **Physicalism**?

Preset *item types* can be declared and used instead of configuring the `\pitem`s one at a time:

```
\begin{prop}
  \pitem[Nihilism,
    type=long]
    There is nothing.
    \label{nihilism}
\end{prop}
Does \ref{nihilism} imply
\ref{phys}, \ref{dual}, or
both?  Discuss.
```

Nihilism There is nothing.

Does Nihilism imply **Physicalism**, (3), or both? Discuss.

Shortcut commands—e.g., `\litem[`⟨*keys*⟩`]` for `\pitem[type=long, `⟨*keys*⟩`]`—can also be defined. The package loads with a range of predefined types.

When the package is loaded with `\usepackage[equations]{propositions}`, first-level `\pitem`s use the same counter as equations. (This looks better with the `leqno` option to `\documentclass`.)

```
\begin{equation}
  \exists x (\text{Mental}(x)
  \wedge \text{Physical}(x))
\end{equation}
\begin{prop}
  \pitem
  There is overlap between
  the mental and the physical.
\end{prop}
```

(4) $\exists x(\text{Mental}(x) \wedge \text{Physical}(x))$

(5) There is overlap between the mental and the physical.

The `\ptag` command (requires `amsmath`) is an analogue of `\pitem` that works inside displayed math environments.

```
\begin{equation}
  \ptag[Monism] \label{mon}
  \exists x \forall y(y = x)
\end{equation}
Is \ref{mon} compatible with
\ref{dual}?
```

Monism $\exists x \forall y(y = x)$

Is **Monism** compatible with (3)?

5 The `prop` environment and `\pitem`

`\begin{prop}[⟨keys⟩]`
  ⟨*environment content*⟩
`\end{prop}`

Creates a displayed list of propositions. It is a standard LaTeX list, so by default its formatting will depend on the standard length parameters like `\itemsep` and `\topsep`, although these can be overridden by setting package keys.

Within `prop`, `\pitem` (see below) creates labelled items. The ordinary `\item` command is still available for unlabelled items.

Any display math environment (`\equation`, `align`, etc.) used inside `prop` or `inlineprop` automatically increments the nesting level for its duration, so that `\ptag` inside such an environment behaves as if it were one level deeper.

The optional ⟨*keys*⟩ argument accepts the same keys as `\propoptions`→P. 11 (section 8), with effects local to this environment.

`\begin{inlineprop}[⟨keys⟩]`
  ⟨*environment content*⟩
`\end{inlineprop}`

Like `prop`, but does not create a list. Allows `\pitem` to be used outside list environments, e.g. for generating numbers at the beginning of paragraphs. Steps the `prop` counter and increments the nesting level. Accepts the same optional ⟨*keys*⟩ as `prop`.

`\pitem[⟨keys⟩]`

Inside the `prop` and `inlineprop` environments, introduces a labelled proposition. The optional argument is a comma-separated list of ⟨*key*⟩=⟨*value*⟩ pairs.

When used without an optional argument (or without setting `name`, `counter`, or `type`), it behaves as `\pitem[type=`⟨*type*⟩`]`, where ⟨*type*⟩ depends on the nesting depth. The defaults are `numbered`, `leveltwo`, `levelthree`, `levelfour`, `levelfive`; these can be changed with the `level` ⟨*n*⟩ keys (section 8).

The following keys can be used in the optional argument of `\pitem`:

`name=`⟨*text*⟩ (no default)

The proposition's name. A bare string (without `=`) in the key list is equivalent to `name=`⟨*text*⟩.

`type=`⟨*type*⟩ (no default)

An item type, equivalent to a preset collection of keys. Types can be declared using `\SetItemType`→P. 7 or `\DeclareNumberedType`→P. 7, and several come predefined (section 6).

`counter=`⟨*name*⟩ (no default)

Counter to use. The counter is automatically stepped, and the item's `name` is set to `\the`⟨*name*⟩, though this can be overridden by explicitly setting `name`.

`align=`⟨*type*⟩ (no default)

How the label should be positioned. Possible values: `default` (offset controlled by `\labelwidth` and `\labelsep`), `right` (right-aligned, like `enumerate`), `center` (centered within the label box), `flush` (aligned with left margin of item text), `nextline` (label on its own line), `left-nextline`, and `flush-nextline`. Has no effect inside `inlineprop` or `\ptag`. For left-margin alignment, use `labelindent=0em` instead.

`labelindent=`⟨*length*⟩ (no default, per-item)

Positions the label's left edge at ⟨*length*⟩ from the enclosing left margin. For example, `labelindent=0em` aligns the label with the surrounding text. When the label plus `\labelsep` would overflow into the item text, the first line is pushed right automatically (like the global `labelindent` key). Can be set per-item, per-type, or globally via `\propoptions`.

`labelwidth=`⟨*length*⟩ (no default, per-item)

Overrides the effective label box width for this item. The label is repositioned as if `\labelwidth` were ⟨*length*⟩, without changing the actual list dimension.

`labelsep=`⟨*length*⟩ (no default, per-item)

Overrides the effective label-to-text gap for this item.

`itemindent=`⟨*length*⟩ (no default, per-item)

Overrides the effective first-line indent for this item. Only increases are fully supported; decreases are clamped to zero.

`format=`⟨*template*⟩ (no default)

Formatting applied to the `name`: use `#1` for the argument, e.g. `format=\textbf{(#1)}`. Shorthand for setting both `display format` and `ref format`.

`name format`=⟨*template*⟩ (no default)

Format for displaying the name in the proposition's label. Does not affect cross-references. (Synonym: `display format`.)

`display format`=⟨*template*⟩ (no default)

Synonym for `name format`.

`label format`=⟨*template*⟩ (no default)

Format applied to the *entire* assembled label, i.e. the name (after `name format`), shorthand, and gloss together. Use `#1` for the whole assembled text. For example, `label format=#1:` appends a colon after the complete label. Does not affect cross-references.

`ref format`=⟨*template*⟩ (no default)

Format for subsequent cross-references to this proposition. Does not affect the display.

`shorthand`=⟨*text*⟩ (no default)

An abbreviation displayed after the name. If present, the shorthand becomes the reference text: `\ref` produces the shorthand (formatted with `ref format`) rather than the full name.

`shorthand format`=⟨*template*⟩ (initially `~[#1]`)

Format for displaying the shorthand in the label.

`gloss`=⟨*text*⟩ (no default)

A parenthetical gloss displayed after the name. Does not affect cross-references.

`gloss format`=⟨*template*⟩ (initially `~(#1)`)

Format for displaying the gloss in the label.

`ref`=⟨*text*⟩ (no default)

Explicitly set the reference text, overriding what would be derived from `name`, `counter`, or `shorthand`.

`label`=⟨*label*⟩ (no default)

Equivalent to a trailing `\label{`⟨*label*⟩`}`.

`crefname`=⟨*type*⟩ (no default)

When cleveref is loaded, assigns an arbitrary reference type to this proposition. For example, `crefname=lemma` causes `\cref` to use the names defined by `\crefname{lemma}{...}{...}` instead of the default `proposition` type. The ⟨*type*⟩ must be known to cleveref; new types can be declared with `\crefname`.

`\ptag[`⟨*keys*⟩`]`

Available only when amsmath is loaded. Works inside displayed math environments like `equation` and `align`. Accepts the same keys as `\pitem`→P. 4, except that `align` has no effect (positioning is controlled by the tag placement system).

6 Item types

\SetItemType{⟨*name*⟩}{⟨*keys*⟩}

Defines or modifies an item type for use with the type key. All \pitem→ P. 4 keys are accepted, plus the following:

macro=⟨*command*⟩ (no default)

A new user macro, equivalent to \pitem[type=⟨*name*⟩]. Any further keys given to the macro are passed to \pitem.

If the type ⟨*name*⟩ already exists, \SetItemType modifies or adds keys. For example, \SetItemType{default}{align=flush} changes the alignment of the built-in default type while preserving its other settings.

```
\SetItemType{angle}{
  labelindent = 0em,
  display format =
    \textbf{$\langle$#1$\rangle$},
  ref format     =
    $\langle$#1$\rangle$,
  macro = \angitem
}
\begin{prop}
  \angitem[Angle thesis]
  Everything is angular.
\end{prop}
No further discussion of
\lastref{} is needed.
```

⟨**Angle thesis**⟩ Everything is angular.

No further discussion of ⟨Angle thesis⟩ is needed.

\DeclareNumberedType{⟨*name*⟩}[⟨*keys*⟩]

Creates a new LaTeX counter named ⟨*name*⟩ and a matching item type with counter=⟨*name*⟩. All \SetItemType keys are accepted, plus:

parent=⟨*counter*⟩ (no default)

A parent counter; the new counter resets when the parent steps (same mechanism as \numberwithin). A dedicated prop counter (stepped by each prop and inlineprop) is available for non-persistent numbering.

counter format=⟨*format*⟩ (default ⟨*name*⟩\arabic{⟨*name*⟩})

The representation of the new counter (\the⟨*name*⟩).

```
\DeclareNumberedType{P}
\begin{prop}
  \pitem[counter=P] First premise.
  \label{p1}
  \pitem[counter=P] Second premise.
  \label{p2}
\end{prop}
From \ref{p1} and \ref{p2}\ldots
```

P1 First premise.

P2 Second premise.

From **P1** and **P2**. . .

6.1 Built-in types

The following item types are predefined. `\SetItemType` can modify their behaviour.

Type	Shortcut	Counter	Display	Ref	Align
`plain`	none	none	Name	Name	left
`default`	none*	none	**Name**	**Name**	left
`long`	`\litem`	none	**Name**	Name	flush
`bullet`	`\bitem`	none	•	•	default
`roman`	`\ritem`	`roman`	(i)	(i)	left
`alph`	`\aitem`	`alph`	(a)	(a)	left
`numbered`	none†	`numpropi`‡	(1)	(1)	left
`leveltwo`	none†	`numpropii`	a.	(1a)	left
`levelthree`	none†	`numpropiii`	(i)	(1a.i)	left
`levelfour`	none†	`numpropiv`	•	•	default
`levelfive`	none†	`numpropv`	–	–	default

* The `default` type is auto-selected when `\pitem` or `\ptag` has an optional argument but no `type` key.

† The `numbered`–`levelfive` types are auto-selected when `\pitem` or `\ptag` has no optional argument, depending on nesting level.

‡ The `equations` package option changes `numbered`'s counter to `equation`. The `numpropii`–`numpropv` counters reset automatically when a `\pitem` at the next lower level is processed.

7 Cross-referencing

Labels placed after `\pitem` items work with the standard `\label`/`\ref` mechanism. The key difference from ordinary LaTeX references is that `\ref` produces *formatted* output: for example, `\textbf` might be applied to the name, or the number might be wrapped in parentheses. The formatting is controlled by the `format` key (or separately by `display format` and `ref format`).

`\Ref{`⟨*label*⟩`}`
`\Ref*{`⟨*label*⟩`}`

Titlecase variant of `\ref`: uppercases the first letter of the formatted output. For example, if `\ref{thesis}` produces 'the Identity Theory', then `\Ref{thesis}` produces 'The Identity Theory'. The starred form suppresses the hyperlink. Note: `\Ref` only affects propositions references (those stored via `\propapply`); for other label types, it behaves like `\ref`.

`\nref{`⟨*label*⟩`}`
`\nref*{`⟨*label*⟩`}`
`\Nref{`⟨*label*⟩`}`
`\Nref*{`⟨*label*⟩`}`

"Naked ref." Outputs the bare reference content with all formatting stripped. If `\ref{premise}` produces '(P1)', then `\nref{premise}` produces 'P1'. The starred form suppresses the hyperlink. `\Nref` is the titlecase variant: it uppercases the first letter of the bare content.

`\nref` is often useful in the argument of `\pitem`, when the the name of one proposition should depend on that of another:

```
\SetItemType{default}{format=(#1)}
\begin{prop}
  \pitem[Phys]
  Everything is physical.
  \label{premise}
  \pitem[\nref{premise}*]
  Almost everything is physical.
  \label{newpremise}
  \pitem[\ref{premise}*]
  The version \ref{newpremise}, which
  uses |\nref|, looks better
  than the one with |\ref|, unless
  for some reason one wants two
  lots of parentheses.
\end{prop}
```

(Phys) Everything is physical.

(7*) Almost everything is physical.

(7*) The version (7*), which uses `\nref`, looks better than the one with `\ref`, unless for some reason one wants two lots of parentheses.

Warning: documents where the name of one item includes a reference to that of another, and there are further references to that item, will require multiple LaTeX runs to resolve all references. To save time, it is better to avoid long chains of dependencies of this sort.

`\oref`[⟨*prefix*⟩][⟨*suffix*⟩]{⟨*label*⟩}
`\oref*`[⟨*prefix*⟩][⟨*suffix*⟩]{⟨*label*⟩}
`\Oref`[⟨*prefix*⟩][⟨*suffix*⟩]{⟨*label*⟩}
`\Oref*`[⟨*prefix*⟩][⟨*suffix*⟩]{⟨*label*⟩}

"Ref with options." Extends `\ref` by injecting a prefix and/or suffix *inside* the formatting. With one optional argument, ⟨*suffix*⟩ is appended; with two, ⟨*prefix*⟩ is prepended and ⟨*suffix*⟩ appended. For instance, if `\ref{premise}` produces '(P1)', then `\oref[*]{premise}` produces '(P1*)' and `\oref[cf.~][*]{premise}` produces '(cf.~P1*)'. The starred form suppresses the hyperlink. `\Oref` is the titlecase variant: it uppercases the first letter of the output.

`\oref` can also be useful in the name of `\pitem`s, if one wants the display format for the modified item to depend on that originally used

```
\begin{prop}
  \label{premise}
  \pitem[name=\oref[*]{premise},
     format=#1]
  This will use boldface and
  parentheses because the original
  referenced item did.
\end{prop}
```

7 This will use boldface and parentheses because the original referenced item did.

Another handy use for `\oref` is in combination with `\nref` to refer to ranges:

```
The first two numbered examples in
this document were
were \oref[--\nref{atomism}]{atoms}.
```

The first two numbered examples in this document were were (1–2).

\lastref[⟨*prefix*⟩]{⟨*suffix*⟩}
\Lastref[⟨*prefix*⟩]{⟨*suffix*⟩}

Formatted reference to the most recently processed `\pitem` or `\ptag`, even without a `\label`. Useful for back-references in running text. With one argument, ⟨*suffix*⟩ is appended; with two, ⟨*prefix*⟩ is also prepended. Use `\lastref{}` for a plain reference. `\Lastref` is the titlecase variant: it uppercases the first letter.

\nLastref

Like `\lastref{}` but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards.

\parentref[⟨*prefix*⟩]{⟨*suffix*⟩}
\Parentref[⟨*prefix*⟩]{⟨*suffix*⟩}

Inside a nested `prop` (or `inlineprop`), produces a formatted reference to the most recent item of the enclosing level. Same argument convention as `\lastref`. `\Parentref` is the titlecase variant: it uppercases the first letter.

\nParentref

Like `\parentref{}` but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards.

`\parentref` and `\nParentref` are useful for making subitems whose names derive from their parent's:

```
\DeclareNumberedType{inner}[
  counter format=\alph{inner},
  display format=#1.]
\begin{prop}
  \pitem[OI] Outer item.
  \label{outer2}
  \begin{prop}
    \pitem[type=inner,
      format=\parentref{.#1}]
    \label{dsub1}
    Ref: \ref{dsub1},
    naked: \nref{dsub1}.
    \pitem[type=inner,
      display format=#1.,
      ref format=\parentref{.#1}]
    \label{dsub2}
    Ref: \ref{dsub2},
    naked: \nref{dsub2}.
  \end{prop}
\end{prop}
```

(OI) Outer item.

(OI.a) Ref: (OI.a), naked: a.

b. Ref: (OI.b), naked: b.

The built-in `leveltwo` and `levelthree` types have `ref format=\parentref{#1}` and `ref format=\parentref{.#1}`, respectively, so that if the parent references as '(**P1**)', a `leveltwo` sub-item references as '(**P1a**)' and `\nref` returns just 'a'.

7.1 How it works: \propapply

\propapply{⟨*template*⟩}{⟨*content*⟩}

Internally, each reference is stored in the `.aux` file as `\propapply{`⟨*template*⟩`}{`⟨*content*⟩`}`. The ⟨*template*⟩ contains formatting with the placeholder `\propfmtarg`→ P. 11 where content appears. At reference time, `\propapply` evaluates the template with `\propfmtarg` bound to ⟨*content*⟩. The `\oref`→ P. 9 and `\nref`→ P. 8 commands work by locally redefining `\propapply`.
In normal use, you need not interact with `\propapply` directly.

`\propfmtarg`

Placeholder used inside templates; expands to the content argument of the enclosing `\propapply`→ P. 10.

8 Package options

`\propoptions{`⟨*keys*⟩`}`

Sets package-level keys. These can also be set:

- In the optional argument of `prop` and `inlineprop` (local to that environment).
- In the preamble with `\usepackage[`⟨*options*⟩`]{propositions}` (global).

Exception 1: keys containing `#` (such as `equation format`) cannot be set in the optional argument of `\usepackage`, due to how LaTeX handles `#` in option values.

Exception 2: `equations` is a global key; it cannot be used in the optional argument of `prop` or `inlineprop`.

8.1 List dimensions

`topsep=`⟨*length / length list*⟩
`partopsep=`⟨*length / length list*⟩
`itemsep=`⟨*length / length list*⟩
`parsep=`⟨*length / length list*⟩
`leftmargin=`⟨*length / length list*⟩
`rightmargin=`⟨*length / length list*⟩
`labelwidth=`⟨*length / length list*⟩
`labelsep=`⟨*length / length list*⟩
`itemindent=`⟨*length / length list*⟩
`listparindent=`⟨*length / length list*⟩

Override the standard LaTeX list dimensions. Accept the same values as `\setlength`, including rubber lengths (e.g. `itemsep=4pt plus 2pt`). Each key may also take a comma-separated list of per-level values: e.g. `leftmargin={2.5em, 0em}`. The first value applies at level 1, the second at level 2, etc. Gaps in the list (e.g. `leftmargin={, 0em}`) cause the class default to be used for that level.

`labelindent=`⟨*length / length list*⟩ (no default)

Positions label left edges at ⟨*length*⟩ from the enclosing margin, adjusting `\labelsep` or `\itemindent` as needed. Positive values move rightward, negative leftward.

`tightspacing` (no value)

Sets all vertical spacing to the compact defaults that the standard document classes use for level-three lists (`topsep` and `itemsep` to `2pt` with stretch/shrink, `parsep` to `0pt`, `partopsep` to `1pt`). Individual dimension keys set afterward override.

`nosep` (no value)

Sets `\topsep`, `\itemsep`, and `\parsep` all to zero.

`continue`=⟨*boolean*⟩ (initially `true`)

When `true` (the default), a `prop` environment that immediately follows another `prop` (with no intervening paragraph text) replaces the normal `\topsep`-based inter-list space with `\itemsep` + `\parsep`, giving the visual appearance of a single continuous list. This works at any nesting level. Set to `false` to suppress this behaviour for a specific environment or globally via `\propoptions`→P. 11.

8.2 Default types

`default type`=⟨*type*⟩ (initially `default`)

The type used when `\pitem` or `\ptag` is given a name or counter but no explicit `type`.

`default ptag type`=⟨*type*⟩ (initially empty)

If set, overrides `default type` for `\ptag` only.

`level n`=⟨*type*⟩ (see below)

Default item type at nesting level n (1–5) when `\pitem` has no optional argument.

Level	**Default type**
1	`numbered`
2	`leveltwo`
3	`levelthree`
4	`levelfour`
5	`levelfive`

The `leveltwo`–`levelfive` types use special counters `numpropii`–`numpropv`, which reset automatically when `pitem` is used at lesser nesting levels. Other types can also use these counters.

8.3 Formatting and referencing equation numbers

`equations` (no value, **global only**)

Shares the `equation` counter between `\pitem[type=numbered]` and standard displayed equations. Also installs the equation format hooks (see below), so that `\oref` and `\nref` work with equation labels.

`equation format`=⟨*template*⟩ (initially `(#1)`)

Shorthand: sets both `equation display format` and `equation ref format`. Also installs the equation format hooks independently of the `equations` option.

`equation display format`=⟨*template*⟩ (initially `(#1)`)

Controls how equation tags appear in the PDF (via `\tagform@`). Use `#1` for the number. Does not affect `\ref` output. Also installs the equation format hooks independently of the `equations` option. Locally scoped.

`equation ref format`=⟨*template*⟩ (initially `(#1)`)

Controls how `\ref` (and `\oref`, `\nref`) render equation labels. Use `#1` for the number. Does not affect the displayed tag. Also installs the equation format hooks independently of the `equations` option. Locally scoped.

9 Compatibility

The propositions package is designed to work with hyperref, cleveref, and amsmath. amsmath is required for `\ptag` and the `equations` option. With cleveref loaded, all `\pitem`s are assigned to a default `proposition` reference type; the `crefname` key can override this.

Recommended load order:

```
\usepackage{hyperref}
\usepackage{amsmath}   % if using \ptag
\usepackage{propositions}
\usepackage{cleveref}  % if used
```

10 Known issues

When using `\ptag` with a named counter (e.g. `\ptag[counter=P]`) inside an amsmath equation environment, hyperref may emit warnings of the form:

```
pdfTeX warning: destination with the same
identifier (name{equation.N}) has been already
used, duplicate ignored
```

These warnings are harmless and do not affect the correctness of cross-references.